

Nurse Profile & Notifications

Frontend Developer Guide

Version: 1.0 | Date: July 2026 | Audience: Frontend developers
Backend: HM-System (FastAPI) | Auth: JWT Bearer token
Modules: /nurse/profile + /nurse/notifications

1. Purpose & Scope

This document describes the Nurse Profile and Nurse Notifications APIs so frontend developers can build the nurse UI correctly. Both modules are fully implemented. Use this guide as the source of truth for endpoints, payloads, validation, permissions, and UI behaviour.

What this covers

- Nurse self-service profile: view, update, upload/delete profile image
- Which fields nurses can edit vs admin-owned (read-only in UI)
- Profile completion rules and image upload constraints
- In-app nurse notifications: list, unread badge, mark one/all as read
- Nurse-relevant notification types, priorities, filters, deep-link hints
- Suggested TypeScript types and frontend checklist

What this does NOT cover

- Clinical nurse APIs (vitals, notes, medications, handovers, alerts) except as notification triggers
- Admin staff APIs (/users) except where they own nurse fields / shift
- Push notifications, WebSocket, or SSE (not implemented - poll unread-count)
- Delete notification or notification preferences (not implemented)

Auth header (all endpoints)

Authorization: Bearer <access_token>

Login via POST /auth/login. JWT must include the required permission. Only role name "nurse" can use profile endpoints. Inactive accounts: 403. Soft-deleted / missing users: 404.

Required JWT permissions (nurse seed)

Permission	Used for
nurse_profile:view	GET /nurse/profile
nurse_profile:update	PUT /nurse/profile
nurse_profile:upload_image	POST /nurse/profile/image
nurse_profile:delete_image	DELETE /nurse/profile/image
notifications:view	List + unread-count
notifications:update	Mark one / mark all read

After seed/permission changes, nurses must re-login so the JWT includes new permissions.

2. Module A - Nurse Profile

Prefix: /nurse | Router: Routers/nurse_profile_router.py | Tag: Nurse Profile

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Only users with role "nurse" can call these endpoints. Admins manage registration_number, employee_id, joining_date, department, and shift via admin staff APIs - not via /nurse/profile.

2.1 Endpoint summary

Method + Path	Status	Permission	Purpose
GET /nurse/profile	200	nurse_profile:view	Load full profile
PUT /nurse/profile	200	nurse_profile:update	Update editable fields
POST /nurse/profile/image	200	nurse_profile:upload_image	Upload avatar
DELETE /nurse/profile/image	200	nurse_profile:delete_image	Remove avatar

2.2 GET /nurse/profile

Request body: none. Returns NurseProfileResponse (nested address, emergency_contact, department, role, shift).

```
{
  "user_id": 21,
  "first_name": "Priya",
  "last_name": "Sharma",
  "email": "priya.nurse@hospital.com",
  "phone": "9876543210",
  "phone_code": "+91",
  "address": {
    "line": "12 Ward Road",
    "city": "Pune",
    "state": "Maharashtra"
  },
  "date_of_birth": "1992-06-15",
  "gender": 2,
  "emergency_contact": {
    "name": "Rahul Sharma",
    "phone": "9123456780"
  },
  "department": { "id": 3, "name": "General Ward" },
  "role": { "id": 4, "name": "nurse" },
  "qualification": "BSc Nursing",
  "registration_number": "NRC-44521",
  "employee_id": "NUR-102",
  "experience_years": 6,
  "joining_date": "2020-03-01",
  "bio": "Ward nurse with focus on postop care.",
  "languages": ["English", "Hindi", "Marathi"],
  "shift": {
    "name": "Morning",
    "start_time": "08:00",
    "end_time": "16:00"
  },
  "profile_image_url": "/uploads/nurse_image/a1b2c3d4.jpg",
  "is_profile_completed": true,
  "profile_completion_percentage": 93,
  "is_active": true,
  "last_login": "2026-07-14T08:01:00+05:30",
  "created_at": "2026-01-10T10:00:00+05:30",
  "updated_at": "2026-07-13T18:20:00+05:30"
}
```

Response field reference:

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Field	Type	Notes
user_id	number	Nurse user id
first_name	string	Read-only
last_name	string null	Read-only
email	string (email)	Read-only
phone	string null	Editable
phone_code	string null	Editable (e.g. +91)
address	object	{ line, city, state } - editable
date_of_birth	string null	YYYY-MM-DD, editable
gender	number null	1-4, see Gender codes
emergency_contact	object	{ name, phone } - editable
department	object null	{ id, name } - admin-owned, read-only
role	object null	{ id, name } - read-only
qualification	string null	Editable
registration_number	string null	Admin-owned, read-only
employee_id	string null	Admin-owned, read-only
experience_years	number null	Editable, 0-60
joining_date	string null	Admin-owned, read-only
bio	string null	Editable
languages	string[]	Editable, default []
shift	object null	{ name, start_time, end_time } - admin-owned
profile_image_url	string null	Relative path under /uploads
is_profile_completed	boolean	Server-computed (see rules below)
profile_completion_percentage	number	0-100, 14 field checks
is_active	boolean	Account status
last_login	datetime null	ISO with timezone
created_at / updated_at	datetime null	Profile timestamps

Gender codes (integer, not label)

- 1 = Male
- 2 = Female
- 3 = Other
- 4 = Prefer not to say

API returns the integer. Map to labels in the UI.

Profile completion rules

- is_profile_completed = true only when qualification AND experience_years (not null) AND bio are all set.
 - profile_completion_percentage: 14 checks - phone, phone_code, address.line, city, state, DOB, gender, emergency name/phone, qualification, experience_years, bio, languages (non-empty), profile image.
- Show a completion progress bar using profile_completion_percentage.

Errors (GET)

- 401 - Missing/invalid JWT
- 403 - Missing permission, inactive account, or role is not nurse

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- 404 - User/profile not found ("Nurse profile not found. Contact admin.")

2.3 PUT /nurse/profile

Content-Type: application/json. Schema: NurseProfileUpdate with extra="forbid". Partial updates are allowed. Empty body / no fields => 400 "No fields to update". Unknown or admin-only fields => 422.

```
{
  "qualification": "BSc Nursing, Critical Care Cert",
  "experience_years": 6,
  "bio": "Ward nurse with focus on postop care.",
  "languages": ["English", "Hindi"],
  "phone": "9876543210",
  "phone_code": "+91",
  "address": {
    "line": "12 Ward Road",
    "city": "Pune",
    "state": "Maharashtra"
  },
  "date_of_birth": "1992-06-15",
  "gender": 2,
  "emergency_contact": {
    "name": "Rahul Sharma",
    "phone": "9123456780"
  }
}
```

Editable vs read-only

Field	Nurse can edit?	UI tip
qualification	Yes	Text, max 255
experience_years	Yes	Number 0-60
bio	Yes	Textarea
languages	Yes	Tag input; trimmed, deduped
phone / phone_code	Yes	max 20 / max 10
address.{line,city,state}	Yes	city/state max 100
date_of_birth	Yes	Date picker YYYY-MM-DD
gender	Yes	Select 1-4
emergency_contact	Yes	name max 120, phone max 20
first_name, last_name, email	No	Display only
registration_number	No	Admin-owned
employee_id	No	Admin-owned
joining_date	No	Admin-owned
department / role	No	Display only
shift	No	Admin updates; nurse gets SHIFT_UPDATED notification
profile_image_url	No in PUT	Use image upload/delete endpoints

Do NOT send name, email, registration_number, employee_id, joining_date, department, role, shift, profile_image_url, is_profile_completed, or profile_completion_percentage in PUT.

Success response: full NurseProfileResponse (same shape as GET).

2.4 POST /nurse/profile/image

multipart/form-data. Field name: file (required).

- Allowed: .jpg .jpeg .png .webp
- Max size: 5 MB
- Empty file rejected
- Stores under uploads/nurse_image/ (or NURSE_PROFILE_UPLOAD_DIR)
- Replaces previous image file on success

```
{
  "message": "Profile image uploaded successfully",
  "profile_image_url": "/uploads/nurse_image/<uuid>.jpg"
}
```

Display image as: API_BASE + profile_image_url (e.g. http://localhost:8000/uploads/nurse_image/...).

2.5 DELETE /nurse/profile/image

```
{
  "message": "Profile image deleted successfully",
  "profile_image_url": null
}
```

- 404 if no image set

Image / update errors

- 400 - Invalid type, empty file, >5 MB, empty PUT body
- 401 / 403 / 404 - same auth/role/profile rules as GET
- 422 - Forbidden extra fields on PUT

3. Module B - Nurse Notifications

Prefix: /nurse/notifications | Router: Routers/nurse_notification_router.py | Tag: Nurse Notifications

Uses the shared notifications table, filtered by the logged-in nurse user_id. Same schema as doctor notifications, different URL prefix and trigger set.

3.1 Endpoint summary

Method + Path	Status	Permission	Purpose
GET /nurse/notifications/unread-count	200	notifications:view	Badge count
GET /nurse/notifications	200	notifications:view	Paginated list
PATCH /nurse/notifications/{id}/read	200	notifications:update	Mark one read
PATCH /nurse/notifications/read-all	200	notifications:update	Mark all read

3.2 GET /nurse/notifications/unread-count

```
{ "count": 3 }
```

Poll every 15-30 seconds for the header badge. No WebSocket/SSE in backend yet.

3.3 GET /nurse/notifications

Query parameters:

Param	Type / rules	Purpose
page	int >= 1, default 1	Page number
limit	int 1-100, default 20	Page size
search	string min 1	ILIKE title, message, created_by_name
is_read	bool optional	Filter unread/read
source_module	enum optional	e.g. NURSE, ADMIN
notification_type	enum optional	e.g. EMERGENCY_ALERT
start_date / end_date	date optional	Inclusive Asia/Kolkata

Sort order: priority CRITICAL then HIGH then NORMAL, then created_at DESC (newest first within priority).

```
GET /nurse/notifications?page=1&limit=20&is_read=false
```

```
{
  "total": 3,
  "page": 1,
  "limit": 20,
  "items": [
    {
      "id": 501,
      "user_id": 21,
      "title": "Handover taken over",
      "message": "Anita Patel took over your handover H0-2026-0012.",
      "notification_type": "HANDOVER_TAKEN_OVER",
      "priority": "HIGH",
      "source_module": "NURSE",
      "reference_type": "HANDOVER",
      "reference_id": 88,
      "created_by": 22,
```

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```
{
  "created_by_name": "Anita Patel",
  "is_read": false,
  "read_at": null,
  "created_at": "2026-07-14T09:15:00+05:30"
}
```

NotificationResponse fields:

Field	Type	Notes
id	number	Notification id
user_id	number	Recipient nurse
title	string	Short headline
message	string null	Body text
notification_type	enum string	See type table
priority	NORMAL HIGH CRITICAL	UI badge styling
source_module	enum string	Which module created it
reference_type	enum string	Deep-link entity type
reference_id	number	Deep-link entity id
created_by	number null	Actor user id
created_by_name	string null	Actor display name
is_read	boolean	Read flag
read_at	datetime null	Set when marked read
created_at	datetime	ISO with timezone

3.4 PATCH /nurse/notifications/{notification_id}/read

Marks one notification as read for the current nurse. Idempotent (already-read returns success). Sets is_read=true and read_at=now(IST). 404 if notification does not belong to this nurse.

Response: full NotificationResponse.

3.5 PATCH /nurse/notifications/read-all

```
{ "message": "All notifications marked as read" }
```

Marks every unread notification for this nurse as read.

4. Nurse notification types (what you will actually receive)

The shared NotificationType enum has many values (including doctor appointment/lab types). Nurses typically only receive the types below. Filter UI and deep-links around these.

notification_type	priority	source	reference	When created
EMERGENCY_ALERT	by severity	NURSE	ALERT	Alert assigned to nurse, or auto HIGH/CRITICAL alert notifies triggering nurse
HANDOVER_TAKEN_OVER	HIGH	NURSE	HANDOVER	Another nurse takes over your submitted handover
SHIFT_UPDATED	HIGH	ADMIN	SCHEDULE	Admin changes shift name / start / end on nurse profile
ADMIN_UPDATE	HIGH	ADMIN	USER	Admin changes department, deactivates, or deletes nurse account

Deep-link suggestions

- reference_type = ALERT + reference_id -> emergency alert detail screen
- reference_type = HANDOVER + reference_id -> handover detail / take-over screen
- reference_type = SCHEDULE -> open profile shift section (read-only)
- reference_type = USER (ADMIN_UPDATE) -> show notice; may force logout if deactivated

Enums (full, for filter dropdowns)

NotificationType: NEW_APPOINTMENT | APPOINTMENT_CANCELLED | APPOINTMENT_RESCHEDULED | PATIENT_CHECKED_IN | LAB_REPORT_READY | LAB_REPORT_UPDATED | PRESCRIPTION_CREATED | PRESCRIPTION_UPDATED | EMERGENCY_ALERT | ADMIN_UPDATE | HANDOVER_TAKEN_OVER | SHIFT_UPDATED

Priority: NORMAL | HIGH | CRITICAL

SourceModule: OPD_BILLING | LAB | RECEPTIONIST | NURSE | PHARMACY | ADMIN | SYSTEM

ReferenceType: APPOINTMENT | LAB_ORDER | PRESCRIPTION | BILL | PATIENT | USER | SCHEDULE | LEAVE | HANDOVER | ALERT

5. Suggested TypeScript types

```
type GenderCode = 1 | 2 | 3 | 4;

interface AddressInfo {
  line?: string | null;
  city?: string | null;
  state?: string | null;
}

interface EmergencyContactInfo {
  name?: string | null;
  phone?: string | null;
}

interface NurseProfile {
  user_id: number;
  first_name: string;
  last_name?: string | null;
  email: string;
  phone?: string | null;
  phone_code?: string | null;
  address: AddressInfo;
  date_of_birth?: string | null;
  gender?: GenderCode | null;
  emergency_contact: EmergencyContactInfo;
  department?: { id: number; name: string } | null;
  role?: { id: number; name: string } | null;
  qualification?: string | null;
  registration_number?: string | null;
  employee_id?: string | null;
  experience_years?: number | null;
  joining_date?: string | null;
  bio?: string | null;
  languages: string[];
  shift?: {
    name?: string | null;
  }
}
```



```

    start_time?: string | null;
    end_time?: string | null;
  } | null;
  profile_image_url?: string | null;
  is_profile_completed: boolean;
  profile_completion_percentage: number;
  is_active: boolean;
  last_login?: string | null;
  created_at?: string | null;
  updated_at?: string | null;
}

interface NurseProfileUpdate {
  qualification?: string;
  experience_years?: number;
  bio?: string;
  languages?: string[];
  phone?: string;
  phone_code?: string;
  address?: AddressInfo;
  date_of_birth?: string;
  gender?: GenderCode;
  emergency_contact?: EmergencyContactInfo;
}

interface NurseNotification {
  id: number;
  user_id: number;
  title: string;
  message?: string | null;
  notification_type: string;
  priority: "NORMAL" | "HIGH" | "CRITICAL";
  source_module: string;
  reference_type: string;
  reference_id: number;
  created_by?: number | null;
  created_by_name?: string | null;
  is_read: boolean;
  read_at?: string | null;
  created_at: string;
}

```

6. Frontend implementation checklist

Profile

- [] Load GET /nurse/profile on Profile page mount
- [] Render nested address / emergency_contact / department / shift
- [] Disable admin-owned fields (registration, employee_id, joining, dept, shift)
- [] PUT only editable fields; reject extra keys (extra=forbid)
- [] Show completion % progress + is_profile_completed banner
- [] Map gender int 1-4 to UI labels
- [] Image upload via multipart field `file`; preview with API_BASE + url
- [] Delete image updates UI to null avatar

Notifications

- [] Poll GET /nurse/notifications/unread-count for header badge
- [] Inbox: list with page/limit; tabs or filter for is_read
- [] Style CRITICAL / HIGH / NORMAL differently
- [] Click item -> PATCH .../read then deep-link by reference_type
- [] Mark all read button -> PATCH .../read-all then refresh count
- [] Handle EMERGENCY_ALERT, HANDOVER_TAKEN_OVER, SHIFT_UPDATED, ADMIN_UPDATE
- [] After admin deactivate/delete notice, force re-auth if 401/403

Common

- [] Always send Authorization: Bearer <token>
- [] Handle 401 (login), 403 (permission/role), 404, 422 validation
- [] Re-login after seed if nurse_profile:* or notifications:* missing in JWT

7. Quick reference - base URLs

Profile:

```
GET    /nurse/profile
PUT    /nurse/profile
POST   /nurse/profile/image    (multipart file)
DELETE /nurse/profile/image
```

Notifications:

```
GET    /nurse/notifications/unread-count
GET    /nurse/notifications
PATCH /nurse/notifications/{notification_id}/read
PATCH /nurse/notifications/read-all
```

Static files:

```
GET    /uploads/nurse_image/<filename>
```

Generated from HM-System-Backend Docs/generate_nurse_profile_notification_pdf.py. Aligned with Routers/nurse_profile_router.py and Routers/nurse_notification_router.py (July 2026).